

GURU CHARAN MAVUDURU

SOC ANALYST | AI-DRIVEN CYBERSECURITY | APPLIED AI & DATA SCIENCE

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PROFESSIONAL SUMMARY

Cybersecurity and applied AI professional pursuing an M.Tech in Artificial Intelligence and Data Science Engineering at IIT Patna. Hands-on experience in SOC concepts, security monitoring, incident investigation, cybersecurity training, Python-based automation, and AI-assisted threat analysis. Built a retrieval-augmented incident-response copilot, a multi-VM SOC lab with SIEM telemetry, and operational automation solutions using Google tools. Targeting SOC Analyst, Cybersecurity Analyst, Security Operations, Threat Detection, or AI-for-Cybersecurity roles.

TECHNICAL SKILLS

Security Operations: SOC operations, alert triage, incident response, log analysis, threat detection, phishing analysis, malware fundamentals, vulnerability assessment, network security, MITRE ATT&CK

SIEM & Security Tools: Splunk, Microsoft Sentinel, IBM QRadar, Elastic Stack, Security Onion, Sysmon, Winlogbeat, Filebeat, Zeek, Wireshark, Nmap, Burp Suite, Kali Linux, Metasploit

AI & Data Science: Python, retrieval-augmented generation, NLP, SentenceTransformers, FAISS, embeddings, similarity search, prompt engineering, Pandas, NumPy, data visualisation

Systems & Development: Windows, Linux, Ubuntu Server, Oracle VirtualBox, Git, GitHub, SQL, HTML, CSS, JavaScript, React, Vite, Express.js, TypeScript, Google Colab, Apps Script

PROFESSIONAL EXPERIENCE

Computer Teacher | Shree Swaminarayan Gurukul International School

Jun 2025 - Present

Tirupati, Andhra Pradesh

- Teach computer fundamentals, Python, HTML, CSS, digital literacy, and safe internet practices through classroom and laboratory sessions.
- Create lesson plans, assessments, demonstrations, and practical exercises tailored to different grade levels and learning abilities.
- Support computer-lab operations, shared digital resources, technical documentation, school presentations, and technology-enabled academic activities.

Cybersecurity Trainer - Part Time | SynthoQuest IT Solutions

Dec 2024 - May 2025

Remote

- Delivered practical training on SOC fundamentals, incident investigation, phishing, malware, network security, vulnerability assessment, and web security.
- Guided learners through hands-on use of Kali Linux, Nmap, Wireshark, Burp Suite, Splunk, Microsoft Sentinel, and IBM QRadar.
- Prepared structured lab exercises, demonstrations, technical notes, and career guidance for graduates and working professionals.

Junior Design Engineer | CADSYS (INDIA) LIMITED

Apr 2025 - Jun 2025

India

- Worked in an engineering design environment, following structured technical workflows, quality checks, documentation practices, and delivery timelines.
- Strengthened analytical problem-solving, professional communication, accuracy, and cross-functional coordination in a production-oriented setting.

SELECTED CYBERSECURITY & AI PROJECTS

CyberGPT: Retrieval-Augmented Security Incident Response Copilot | Independent Research Project

2026

- Built a Python-based SOC copilot that classifies incident narratives and retrieves relevant containment, investigation, recovery, and prevention playbooks.
- Created a structured knowledge base covering approximately 30 attack categories and mapped incidents to MITRE ATT&CK techniques, severity, indicators, affected assets, and recommended actions.
- Implemented SentenceTransformers embeddings, FAISS similarity search, vector normalisation, hybrid keyword boosting, novelty detection, automated SOC reports, and a review queue.

SOC Home Lab and Security Monitoring Dashboard | Independent Project

2026

- Designed a four-system virtual SOC environment using Kali Linux, Windows 10, Ubuntu Server, and Splunk with isolated internal networking and NAT for updates.
- Configured Windows auditing, Sysmon, PowerShell logging, and Winlogbeat forwarding to investigate failed logons, reconnaissance, suspicious execution, and endpoint activity.
- Developed a browser-based SOC dashboard for alert severity, incident status, investigation notes, analyst workflow, and reporting.

SELECTED PROJECTS - CONTINUED

Sahaaya360 / SahaayaOS Lite | Cloud Operations Automation Project 2026

- Designed a low-cost institutional operations platform for schools, hostels, apartments, clinics, hospitals, colleges, and small organisations using Google Sheets, Forms, Apps Script, and Python.
- Created modules for tickets, assets, vendors, safety checks, cyber incidents, DPDP readiness, institutions, and monthly reports, with automated IDs, classification, priority, SLA, ownership, and status tracking.
- Built backup workflows, operational dashboards, project tracking, pitch materials, and lead-management resources for an MVP-oriented product concept.

Osprey-Based MPPT for Solar PV Under Partial Shading | B.Tech Research Project 2024 - 2025

- Modelled a solar PV system and boost converter in MATLAB/Simulink and compared PSO, Osprey Optimisation, and Modified Osprey Optimisation for maximum-power-point tracking.
- Evaluated convergence, output power, stability, and partial-shading performance; experimental results showed approximately 97% tracking efficiency for Osprey-based approaches.
- Prepared simulation datasets, technical graphs, comparative analysis, and research documentation.

ADDITIONAL DIGITAL PROJECTS

Personal Portfolio and Technical Blog Platform | GitHub Pages Brand Project 2026

- Upgraded a personal portfolio to present cybersecurity, AI, data science, engineering, teaching, and research work through structured project case studies and direct social links.
- Designed a blog workflow in which technical documents are displayed as titled cards with concise descriptions and direct open-and-read actions, supporting long-term personal branding and knowledge sharing.

Mavudurus Dynamic Wedding Invitation Platform | Responsive Web Project 2026

- Developed a private, responsive digital invitation experience with event sections, family galleries, cover images, theme controls, and simplified navigation for mobile and desktop users.
- Organised Drive-hosted media and content into maintainable folders and improved usability by removing unnecessary developer-facing elements and refining the guest journey.

ADDITIONAL TECHNICAL EXPERIENCE

Electrical Engineering Intern | APTRANSCO - 220/132 kV Substation May 2024 - Jun 2024

Simhachalam, Andhra Pradesh

- Observed power-transmission operations, transformers, switchgear, protection relays, control panels, busbar arrangements, and electrical safety procedures.
- Documented technical observations and gained exposure to preventive maintenance, system monitoring, and structured operational practices in a critical-infrastructure environment.

EDUCATION

Master of Technology - Artificial Intelligence and Data Science Engineering | Indian Institute of Technology Patna Apr 2026 - Present

Pursuing | Focus: Artificial Intelligence, Machine Learning, Data Science, Generative AI, Data Engineering, and applied research

Bachelor of Technology - Electrical and Electronics Engineering | Aditya Institute of Technology and Management 2021 - 2025

CGPA: 7.8/10 | Relevant work: programming, MATLAB/Simulink, power electronics, renewable energy systems, and engineering mathematics

Intermediate | Gurajada Educational Society 2021

88.4%

Secondary School Certificate | Gurajada Educational Society 2019

CGPA: 9.3/10

CERTIFICATIONS & PROFESSIONAL TRAINING

- AICTE-EduSkills Palo Alto Networks Cybersecurity Virtual Internship
- AWS Academy Cloud Foundations and APSSDC AWS Cloud Computing Training
- Tata Group Cybersecurity Analyst Job Simulation
- Introduction to Cybersecurity, Security Operations Center, and Introduction to Generative AI

LEADERSHIP & ADDITIONAL VALUE

- Led an eight-member team for an APSSDC AWS seminar and coordinated the preparation and presentation of technical concepts.
- Create technical documentation, lab guides, cybersecurity awareness content, project reports, dashboards, and portfolio material for diverse audiences.
- Research interests: AI-driven cybersecurity, SOC automation, threat detection, incident response, explainable AI, generative AI, and security engineering.